Grocery Sales Dashboard – Power BI Project Documentation

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Table of Contents

- Introduction
- Data Source and Description
- Data Cleaning and Transformation
- Data Model
- DAX Calculations
- Visuals per Page
- - Home (Overview)
- - Sales & Product
- - Customer Insights
- - Regional & Team Performance
- - Product Insights
- Filters/Slicers Used
- Key Insights
- Exporting and Sharing
- Conclusion
- Appendix

Introduction

This document provides complete project documentation for the Grocery Sales Dashboard developed in Power BI.

The dashboard analyzes a simulated grocery sales dataset to highlight sales performance, product trends, customer behavior, and regional comparisons. The report is designed with five interactive pages (Home, Sales & Product, Customer Insights, Regional & Team Performance, and Product Insights) to allow stakeholders to explore data by date, product, customer, city, and employee.

Data Source and Description

The analysis uses the Grocery Sales Database from Kaggle (https://www.kaggle.com/datasets/andrexibiza/grocery-sales-dataset), which contains simulated sales from 2018-01-01 to 2018-05-09. The dataset includes:

- Sales table (gives sales information about the product which was sold),

- Product table (gives information about the products),

- Employees table (gives information about the employees),

- Customers table (gives the name, address and id of the customer),

- Countries table (gives the country name, country code and its ID),

- Cities table (gives the city name, city id, zip code and the ID of the country it is in),

- Categories table (gives information on the different categories of products sold along with their category ID).

Relationships are defined using key fields like ProductID, CustomerID, etc.

Data Cleaning and Transformation

Performed in Power Query:

- Merged Sales and Product tables on ProductID and named it Product Sales

- Renamed the Product table to products (2)

- Added a new column to the Customers table called Full Name

- Added a new column to the Employees table called Full Name

- Created calculated fields like Total Cost = (Quantity × Product Cost) * (1-Discount)

- Cleaned nulls, standardized data types and formats.

Data Model

Designed using a star schema:

- Background designed using Canva

- Product Sales table at the center

- Related to Product, Customer, and Employee dimension tables

- Enables filters and slicers to interact with sales data seamlessly

DAX Calculations

Key measures created in Power BI:

- Average Purchase = SUM ('Product Sales'[Total Cost])/DISTINCTCOUNT (customers [CustomerID])

- Unique_Customers = DISTINCTCOUNT ('Product Sales'[CustomerID])

- Average Sales by an Employee= SUM('Product Sales'[Total Cost])/DISTINCTCOUNT(employees[EmployeeID])

- Average Price = SUM ('Product Sales'[Total Cost])/SUM ('Product Sales'[Quantity])

- Average_Transaction_Value = SUM ('Product Sales'[Total Cost])/COUNT (sales [SalesID])

Visuals per Page

Home (Overview)

- Cards (Total Sales Revenue,

  Total Transactions,

  Number of Unique Customers,

  Average Transaction Value)

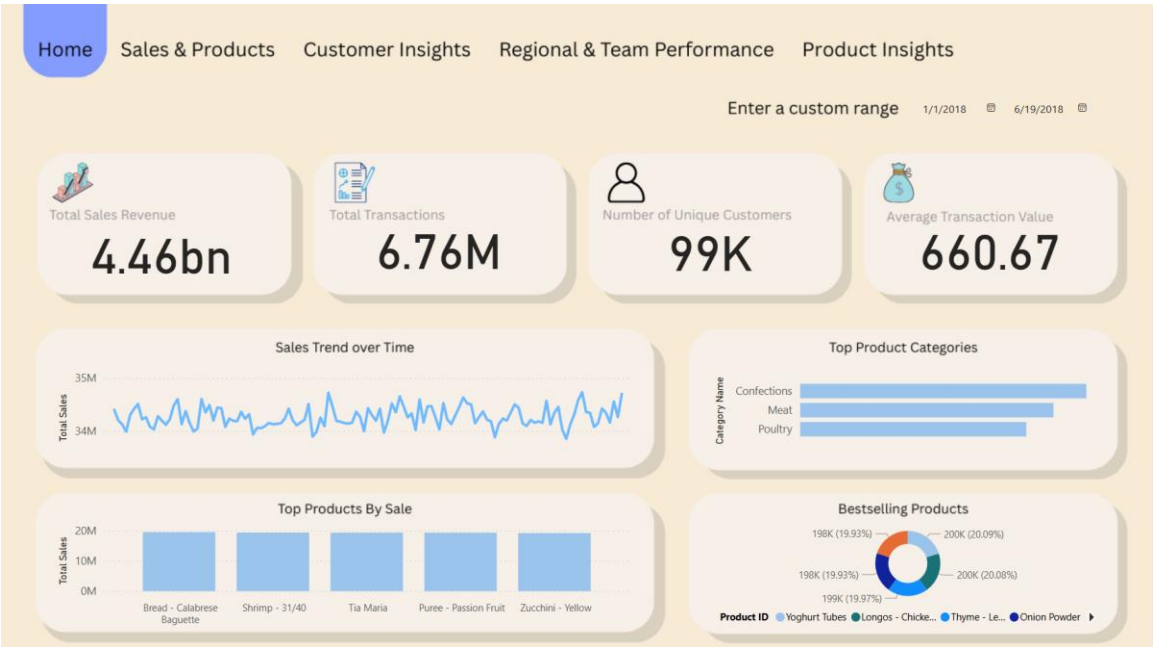
- Sales Over Time (Line Chart)

- Top Product Categories (Horizontal Bar chart)

- Top Products by Sale (Bar chart)

- Bestselling Products (Donut chart)

- Date Slicer



Sales & Product

- Card (Sales by Product Category)

- Sales & Quality Trend (Line Chart)

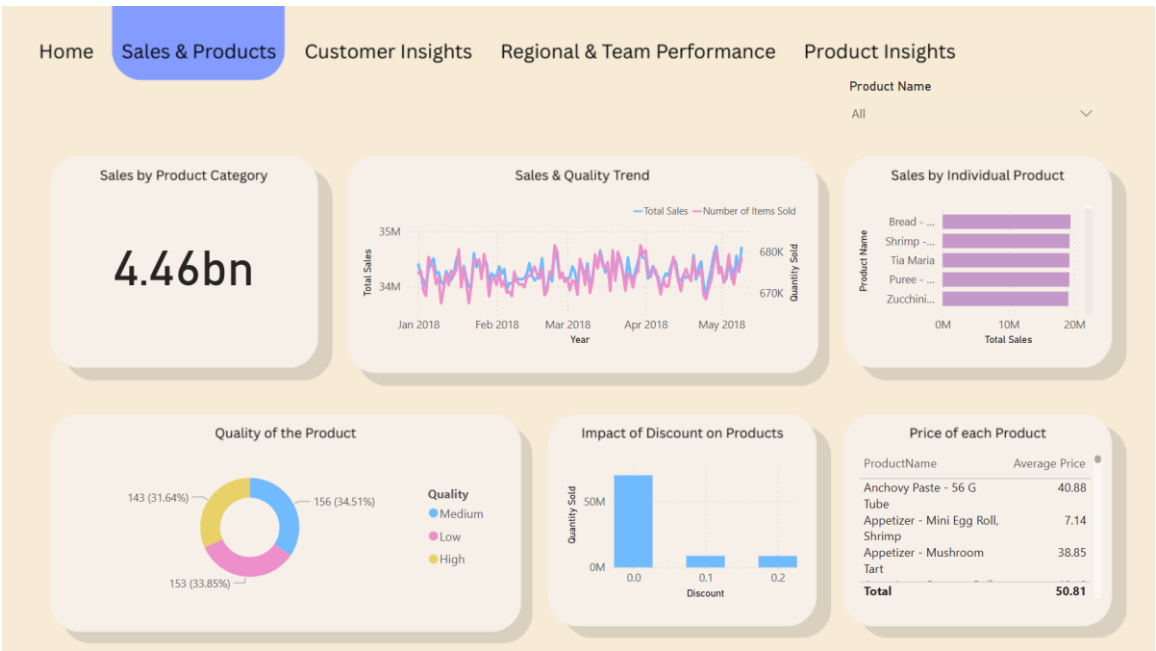
- Sales by Individual Product (Horizontal Bar chart)

- Quality of the Product (Donut chart)

- Impact of Discount on Products (Bar chart)

- Price of each Product (Table)

- Product Name Slicer



Customer Insights

- Map of Customers

- Card (Average Purchase by a Customer)

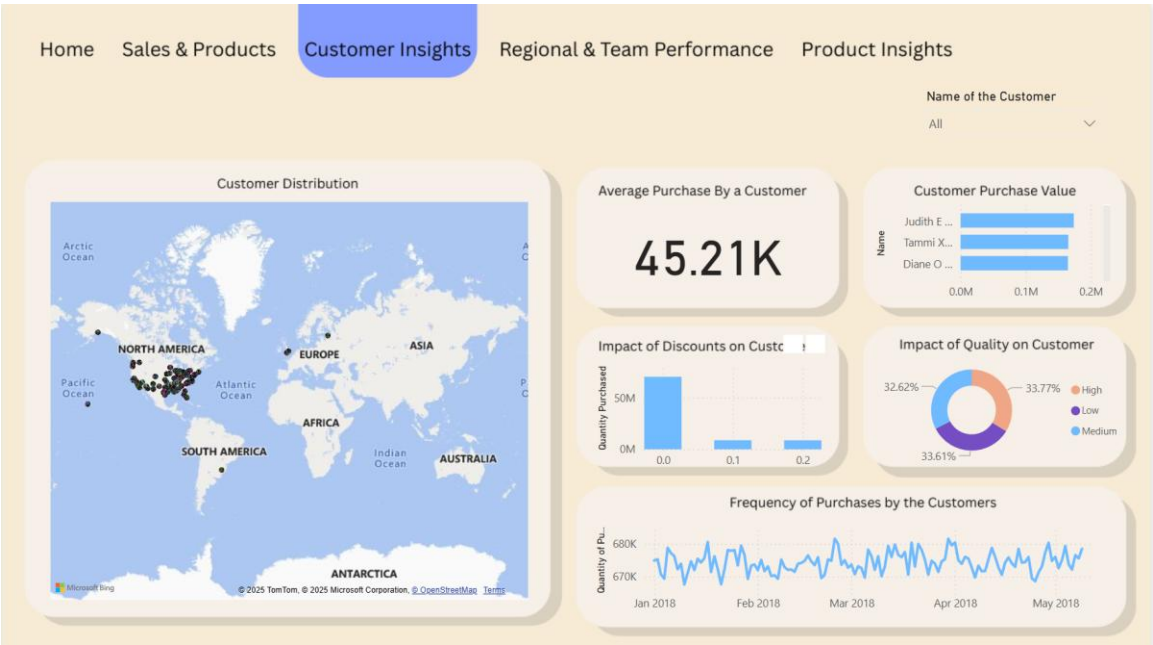
- Customer Purchase Value (Horizontal Bar chart)

- Impact of Discounts on Customers (Bar chart)

- Impact of Quality on Customer (Donut chart)

- Frequency of Purchases by the Customers (Line Chart)

- Customer Name Slicer



Regional & Team Performance

- Card (Average Sales by an Employee)

- Employee Contribution to Sales (Donut chart)

- Top Performing Employees (Bar chart)

- Sales Trend (Line chart)

- Map of Sales Revenue by Region

- Regional and Employee Slicers

Product Insights

- Card (Number of Sales)

- Effect of Price on Sales (Line chart)

- Least Performing Products (Horizontal Bar chart)

- Quantity of Products which cause Allergies (Bar chart)

- Sales of each Category over Time (Area chart)

- Product Slicer

Filters/Slicers Used

- Date Slicer (Home)

- Product Name Slicer (Sales & Product, Product Insights)

- Customer Name Slicer (Customer Insights)

- City & Employee Name Slicer (Regional & Team Performance)

Key Insights

- Sales trends and peak periods identified

- Top-selling and overstocked products highlighted

- Customer purchasing patterns revealed by geography and frequency

- Regional and employee performance compared

- Product-level profitability tracked

Exporting and Sharing

- Export to PDF using Power BI Desktop or Service

- Share via Power BI Service (workspace or app)

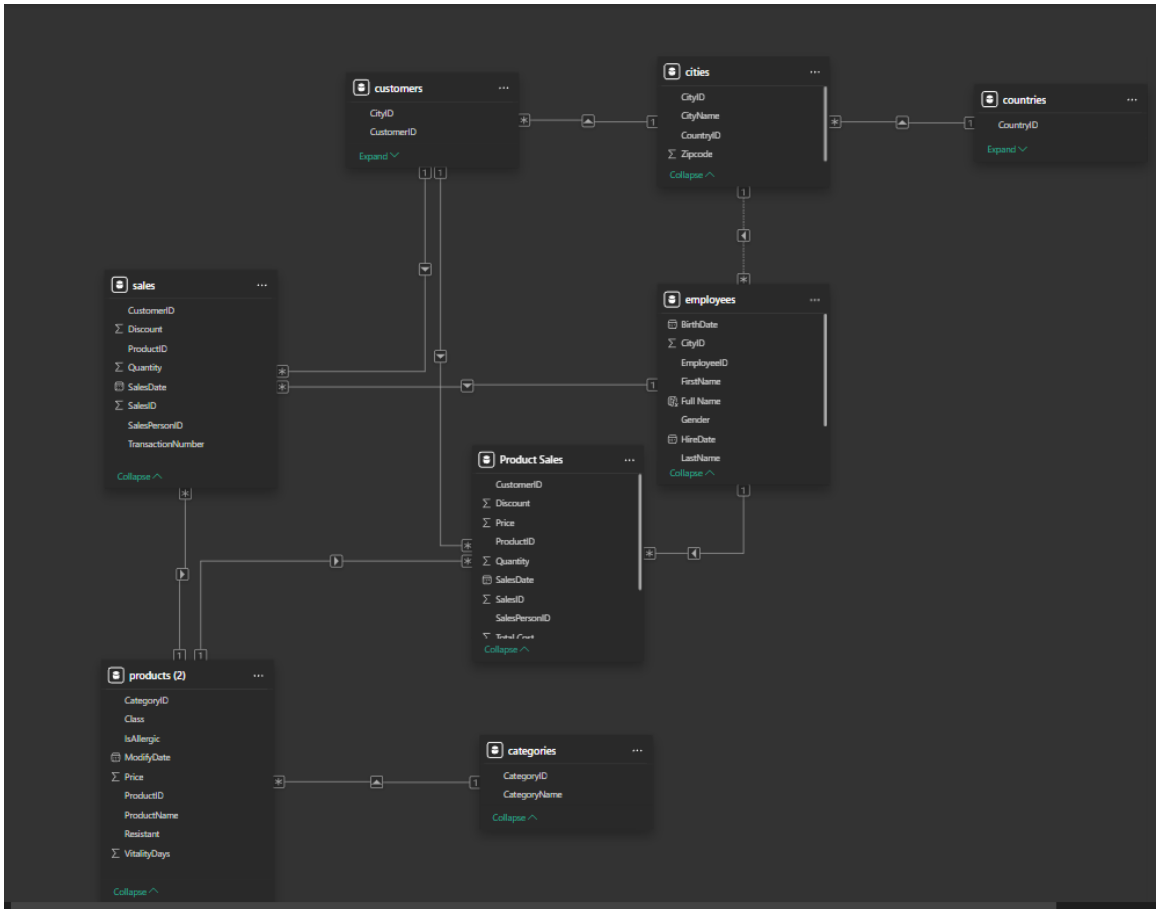
- Embed in Teams or publish to web if permitted

Conclusion

The Grocery Sales Dashboard offers actionable insights through five report pages. It showcases Power BI's capabilities in merging, modeling, calculating, and visualizing sales data interactively and effectively.

Appendix

- Additional Visuals:



From: table (column)	Relationship	To: table (column)	Status
cities (CountryID)	* ◀ 1	countries (CountryID)	Active
customers (CityID)	* ◀ 1	cities (CityID)	Active
∑ employees (CityID)	* ◀ 1	cities (CityID)	Inactive
Product Sales (CustomerID)	* ◀ 1	customers (CustomerID)	Active
Product Sales (ProductID)	* ◀ 1	products (2) (ProductID)	Active
Product Sales (SalesPersonID)	* ◀ 1	employees (EmployeeID)	Active
products (2) (CategoryID)	* ◀ 1	categories (CategoryID)	Active
sales (CustomerID)	* ◀ 1	customers (CustomerID)	Active
sales (ProductID)	* ◀ 1	products (2) (ProductID)	Active
sales (SalesPersonID)	* ◀ 1	employees (EmployeeID)	Active

- Additional DAX Formulas:

- SUM ('Product Sales' [Quantity])
- SUM ('Product Sales' [Total Cost])
- COUNT (‘products (2)’ [Product ID])
- SUM (‘sales’ [Quantity])
- COUNT (‘products (2)’ [ProductName])
- COUNT ('Product Sales' [SalesID])